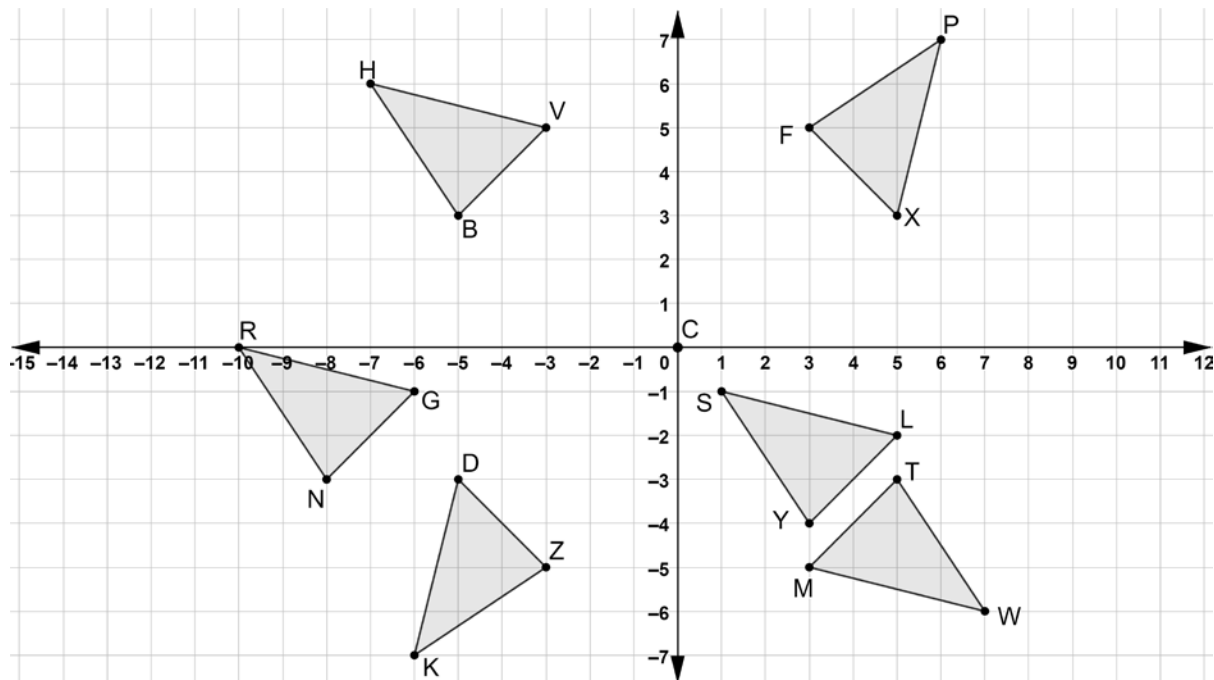


Geometry
Unit 4
Lesson 1 Practice

Name _____
Date _____
Period _____

Six triangles are shown on the coordinate grid.



1. Draw the mapping that shows the translation if $\triangle SYL$ is the preimage and $\triangle HBV$ is the image.
2. Draw the mapping that shows the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.
3. Describe the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.

4. Complete the statements.

The x -values of $\triangle RNG$, the image, can be found by

- ☐ adding
☐ subtracting

___ unit(s) to the x -values of $\triangle SYL$, the preimage. The y -values

of $\triangle RNG$, the image, can be found by

- ☐ adding
☐ subtracting

___ unit(s)

to the y -values of $\triangle SYL$, the preimage.

5. On the coordinate grid draw angle DCV .
6. The algebraic description of the rotation of $\triangle KZD$, the preimage, onto $\triangle HBV$ is the image, is $(x, y) \rightarrow (y, -x)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.
7. The algebraic description of the rotation of $\triangle TWM$, the preimage, onto $\triangle BHV$ is the image, is $(x, y) \rightarrow (-x, -y)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.
8. Describe how the measure of $\angle BCF$ can be used to determine the degrees of rotation where $\triangle PFX$ is the preimage and $\triangle HBV$ is the image.
9. Describe how the coordinates of the vertices of $\triangle PFX$, the preimage, and the coordinates of $\triangle HBV$, the image can be used to determine the degrees of rotation.
10. Describe how the orientation of the triangles helps to determine which transformations were a translation and which transformations were a rotation.